Evaluating and Reducing Hallucinations in LLM-Based Medical Report Summarization

Second Progress Report, Fall 2026 (COMP 698C)

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California State University, Northridge, November 06, 2026

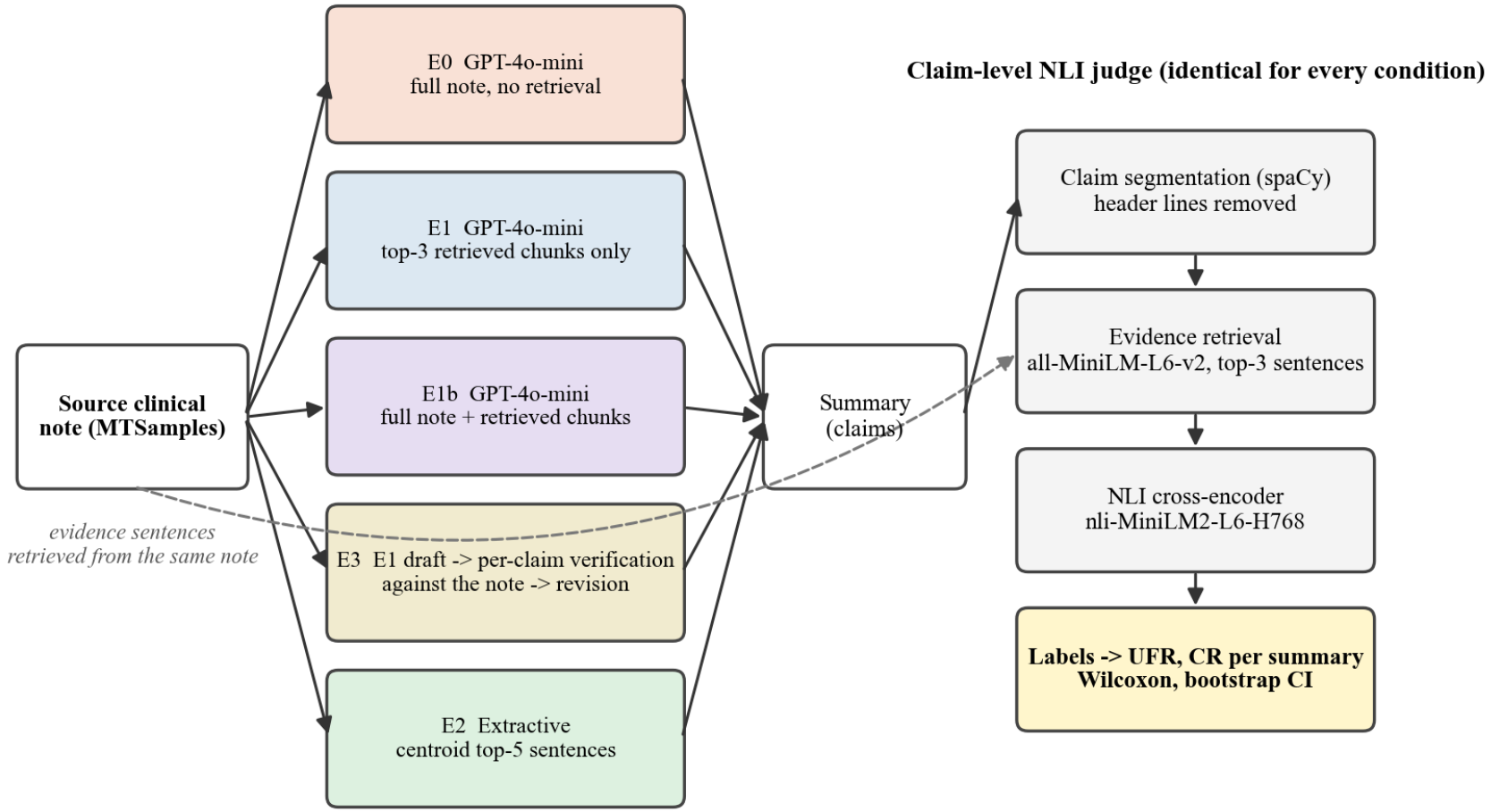
Goal and research questions (from the proposal)

- Goal: measure hallucinations in LLM-generated patient-facing summaries at the claim level with a judge validated against medical experts, and test whether retrieval grounding and verification reduce them on real clinical notes
- RQ1: how often are LLM summary statements unsupported by, or contradicting, the source note?
- RQ2: which statements are unsupported, and how well does the automatic judge agree with medical experts?
- RQ3: do retrieval-augmented generation and verification reduce unsupported and contradicted statements, compared with an extractive reference, and at what cost in coverage?
- Two tasks in the proposal: summarization (main task) and document-grounded question answering (piloted with abstention)
- Three studies: pilot on public MTSamples notes with GPT-4o-mini; judge selection against expert labels; main study on MIMIC-IV hospital courses with a locally run open-weight model

Proposal commitments and where they stand

Commitment in the proposal	Status
Claim-level NLI judge calibrated against expert labels	Done: pilot judge validated (kappa 0.06); six candidates compared; DeBERTa-v3-large NLI selected (kappa 0.33, AUROC 0.78); MedNLI fine-tune: dev accuracy 87.8%
Experiments on MIMIC data	Done: 110 MIMIC-IV hospital courses, Qwen2.5-7B-Instruct served locally, no protected text leaves the laptop
RAG, verification and an extractive baseline	Done in both studies: E0 full-note, E1 RAG (excerpts), E1b RAG (note + excerpts), E3 RAG + Chain-of-Verification, E2 extractive
Coverage against a reference	Done: share of the clinician-written discharge-instruction sentences that the generated summary supports (judge run in reverse)
Citation accuracy	Done: retrieval conditions cite an excerpt after every sentence; the judge checks each citation
Retrieval ablations (chunk size, top-k, BM25 vs dense, citations)	Done: six E1 variants on the same documents
Question answering with abstention	Pilot: three questions per course (medications, follow-up, warning signs), full course vs retrieved excerpts, 'Not stated in the note.' allowed

Pipeline (identical judge for every condition)



Every summary is scored by the same claim-level NLI judge; evidence comes from the same source document.

Data status

- MTSamples (public): 4,999 transcriptions, 624 eligible consult and discharge notes, 50 sampled (seed 42); 2,016 pilot claims labeled
- ann-pt-summ v1.0.1 (PhysioNet, DUA signed): 210 expert-annotated patient summaries with 423 unsupported-fact spans for judge validation and selection; its 110 MIMIC-IV hospital courses with clinician-written discharge instructions are the main-study corpus
- MedNLI v1.0.0 (PhysioNet, DUA signed): 14,049 clinician-written premise and hypothesis pairs from MIMIC-III, used to adapt the judge
- MIMIC-IV-Note v2.2 (PhysioNet, credentialed): downloaded and verified in May 2026; the source of the hospital courses
- Ethics: CITI 'Data or Specimens Only Research' and 'Conflicts of Interest' completed April 10, 2026; every credentialed text is processed by models running on the author's laptop; nothing is sent to an API or to cloud storage

Pilot study: five conditions on 50 MTSamples notes (GPT-4o-mini, MiniLM judge)

Metric	E0 full-note	E1 RAG excerpts	E1b RAG + note	E3 RAG + CoVe	E2 extractive
UFR mean	0.811	0.807	0.759	0.688	0.122
CR mean	0.172	0.130	0.148	0.164	0.096
Supported share of claims	19%	20%	24%	33%	89%
Claims per summary	10.4	8.9	10.2	5.9	4.8
Words per summary	182	156	180	120	69
Coverage of source sentences (cos ≥ 0.6)	16.7%	13.4%	18.5%	16.6%	18.3%

Paired Wilcoxon versus E0: E1 CR $p = 0.011$ (−24%), E1 UFR $p = 0.977$; E1b UFR $p = 0.020$; E3 UFR $p < 0.001$ (−15%; versus E1 $p < 0.001$), E3 CR $p = 0.913$; E2 UFR $p < 0.001$. Verification lowers unsupported facts the most; retrieval alone lowers contradictions modestly; the extractive reference shows the judge errs even on verbatim text. Header lines and abstentions excluded.

Pilot headline: the small MiniLM judge disagrees with medical experts

Group	Sentences	Expert-flagged	Judge-flagged (UFR)	Precision	Recall	Kappa	AUROC
all	1,781	20.4%	84.1%	0.23	0.94	0.06	0.65
generated	805	12.5%	83.7%	0.14	0.96	0.04	0.68
doctor written	976	26.8%	84.3%	0.30	0.94	0.07	0.66
llama 70b original	159	26.4%	83.0%	0.30	0.95	0.10	0.67
gpt4 zero shot	151	6.0%	87.4%	0.06	0.89	0.00	0.50

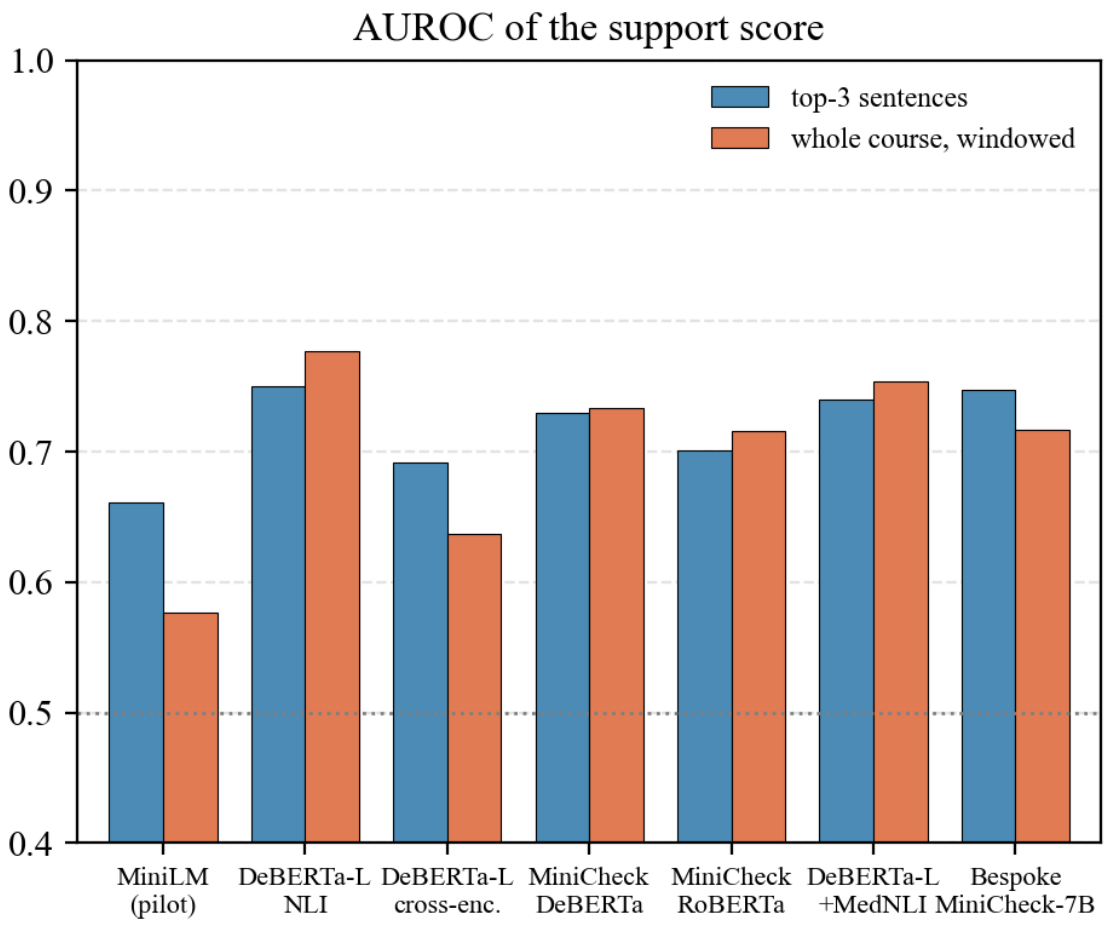
On 1,781 expert-annotated sentences the pilot judge flags 84% where experts flag 20%; kappa 0.06 is chance level; no aggregation variant raised kappa above 0.14. This motivated the judge selection study.

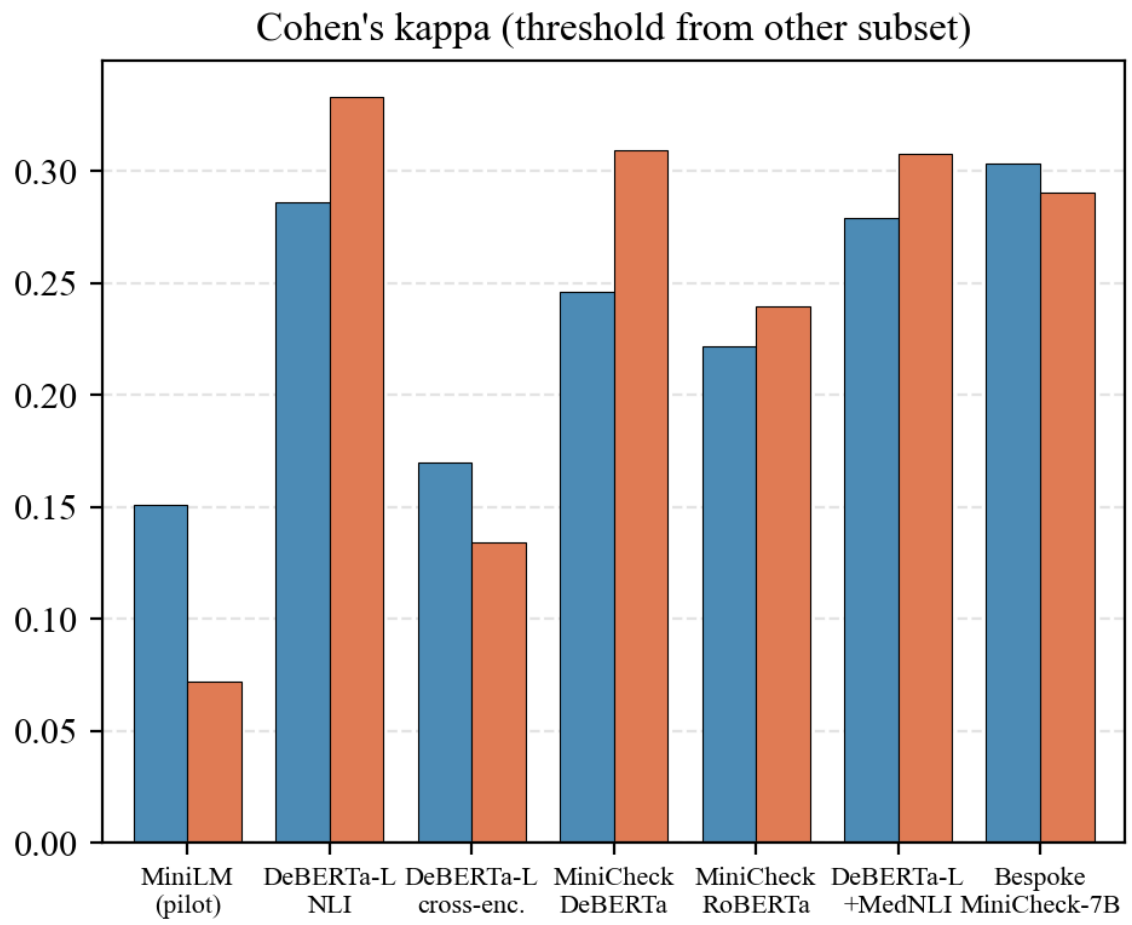
Judge selection: candidates scored against the expert annotations (1,781 sentences)

Judge	Evidence	AUROC	Precision	Recall	Kappa	Flag rate
DeBERTa-v3-large NLI	whole course, windowed	0.78	0.44	0.54	0.33	25%
MiniCheck DeBERTa-v3-large	whole course, windowed	0.73	0.41	0.57	0.31	28%
DeBERTa-v3-large NLI fine-tuned on MedNLI	whole course, windowed	0.75	0.44	0.46	0.31	21%
Bespoke-MiniCheck-7B	top-3 sentences	0.75	0.39	0.60	0.30	31%
Bespoke-MiniCheck-7B	whole course, windowed	0.72	0.38	0.62	0.29	33%
DeBERTa-v3-large NLI	top-3 sentences	0.75	0.36	0.74	0.29	42%
DeBERTa-v3-large NLI fine-tuned on MedNLI	top-3 sentences	0.74	0.41	0.48	0.28	24%
MiniCheck DeBERTa-v3-large	top-3 sentences	0.73	0.34	0.68	0.25	41%
MiniCheck RoBERTa-large	whole course, windowed	0.72	0.41	0.36	0.24	18%
MiniCheck RoBERTa-large	top-3 sentences	0.70	0.33	0.60	0.22	37%
DeBERTa-v3-large cross-encoder NLI	top-3 sentences	0.69	0.28	0.85	0.17	62%
MiniLM cross-encoder	top-3 sentences	0.66	0.28	0.71	0.15	52%
DeBERTa-v3-large cross-encoder NLI	whole course, windowed	0.64	0.27	0.60	0.13	44%
MiniLM cross-encoder	whole course, windowed	0.58	0.24	0.68	0.07	58%

Decision rule fixed in advance: highest kappa on all sentences, threshold chosen on the other subset. Selected: DeBERTa-v3-large NLI (whole course, windowed), kappa 0.33, AUROC 0.78, precision 0.44, recall 0.54: moderate agreement, so main-study rates are estimates from an instrument with a known error profile, and the paired comparisons carry the weight.

Judge selection: AUROC and kappa by candidate and evidence mode





Model size and training data matter more than the aggregation rule; whole-course evidence helps the large models.

Main study design: MIMIC-IV hospital courses

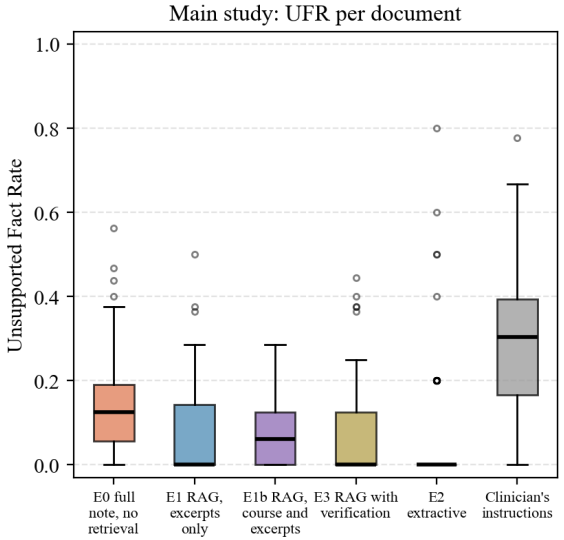
- 110 de-identified Brief Hospital Course sections (ann-pt-summ), each paired with the discharge instructions the treating clinician wrote
- Generator: Qwen2.5-7B-Instruct, 4-bit, served on the laptop with Apple MLX through an OpenAI-compatible interface; the pilot pipeline runs unchanged and no protected text leaves the machine
- Same five conditions; retrieval conditions must cite the excerpt behind every sentence; 'Not stated in the note.' allowed for unsupported sections
- Judge: DeBERTa-v3-large NLI with whole-course evidence; UFR and CR per summary; the doctor-written instructions scored as a human reference
- Coverage measured against the clinician's own instructions (judge in reverse); citation accuracy; six retrieval ablations (3- or 8-sentence chunks, top-2 or top-5, BM25, no citations)
- Question-answering pilot: medications, follow-up and warning signs; full course versus three retrieved chunks; abstention rate, UFR/CR, and support by the clinician's instructions

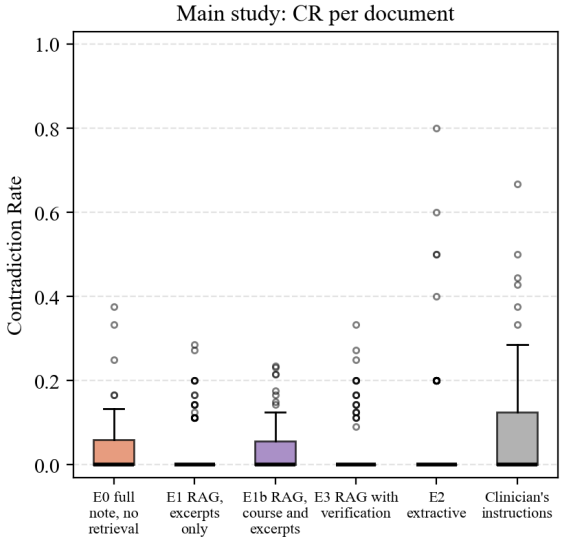
Main study results (n = 110 hospital courses; judge: DeBERTa-v3-large NLI, τ = 0.20)

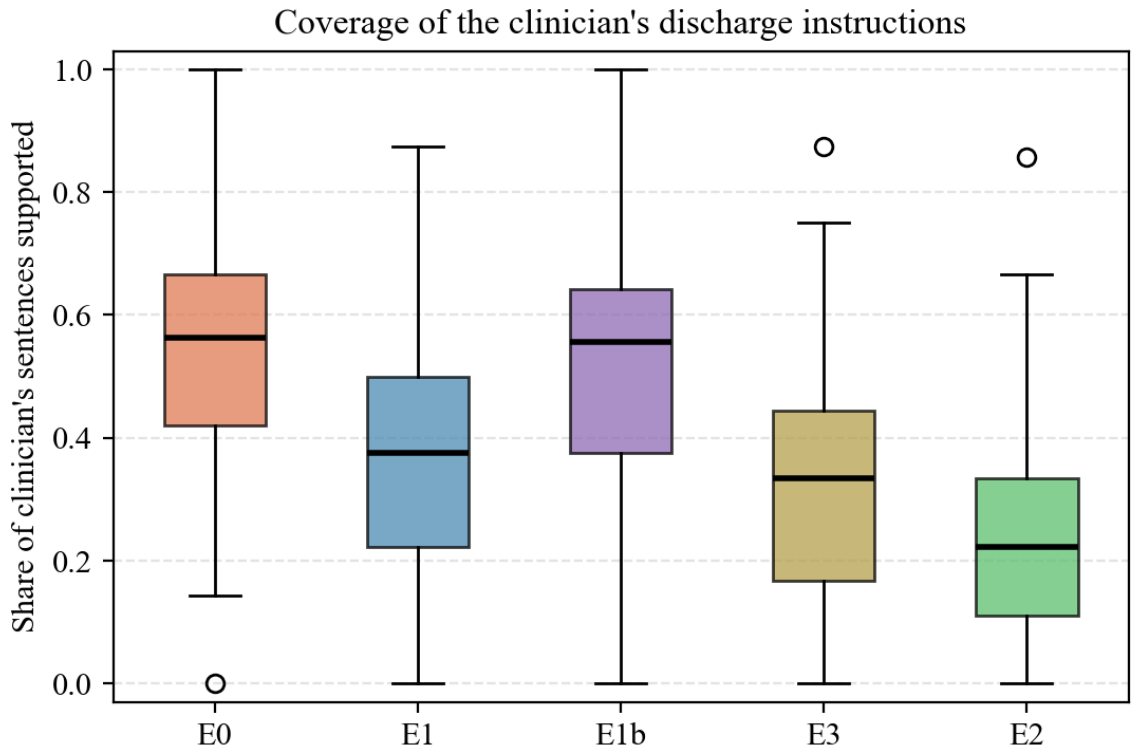
Metric	E0 full-note	E1 RAG excerpts	E1b RAG + course	E3 RAG + CoVe	E2 extractive	Clinician-written
UFR mean	0.137	0.075	0.073	0.068	0.058	0.289
CR mean	0.031	0.030	0.032	0.031	0.053	0.080
Coverage of the clinician's sentences	54.3%	36.9%	53.2%	33.6%	22.5%	—
Citation accuracy	—	84.9%	54.8%	—	—	—
Abstentions per summary	0.0	0.0	0.0	0.7	0.0	0.0
Claims per summary	16.0	8.4	13.8	7.9	4.9	8.9
Words per summary	261	151	228	150	87	133

E1 vs E0: UFR $p < 0.001$, CR $p = 0.985$, coverage $p < 0.001$. E1b vs E0: UFR $p < 0.001$, CR $p = 0.462$, coverage $p = 0.449$. E3 vs E0: UFR $p < 0.001$, CR $p = 0.878$, coverage $p < 0.001$. E2 vs E0: UFR $p < 0.001$, CR $p = 0.169$, coverage $p < 0.001$. Paired two-sided Wilcoxon tests on the same documents.

Main study: per-document distributions







Main study: what the results say

- Every grounded condition cut the unsupported fact rate by about 47 percent versus full-note (E1 0.075, E1b 0.073, E3 0.068 versus E0 0.137; all $p < 0.001$); the three do not differ from one another; CR did not change
- Omission separates them: excerpt-only RAG covered 37% of the clinician's sentences and verification 34%, against 54% for E0; RAG with the whole course kept 53% ($p = 0.449$ versus E0)
- Verbatim extraction scores UFR 0.058: judge error on text supported by construction, a reference rather than a floor; the grounded LLM conditions sit within a few hundredths of it
- The clinicians' own instructions score UFR 0.289 under the same judge: support by the hospital course measures source-faithfulness, not clinical correctness
- Citations: 85% accurate for excerpt-only RAG but 55% for RAG with the whole course; no retrieval setting changed faithfulness; coverage followed the amount of retrieved text
- Question answering: abstention 23% (full course) versus 33% (excerpts), highest for warning signs, which the course rarely states; retrieval lowers unsupported answers partly by abstaining more
- Second judge (Bespoke-MiniCheck-7B, a 7B LLM checker): same ordering of conditions; flags 0.4% of verbatim extracts, so the NLI judge's 0.058 on verbatim text is its own error; confirms 40% of the selected judge's flags; conservative (both judges) and inclusive (either judge) flagging rules give 0.008 to 0.194 for the grounded conditions, alternative estimates rather than bounds

Retrieval ablations: paired difference from the E1 base configuration

Variant	Δ UFR (p)	Δ CR (p)	Δ coverage (p)	Δ citation accuracy (p)
3-sentence chunks	+0.000 (p = 0.902)	+0.013 (p = 0.146)	-0.072 (p < 0.001)	-0.020 (p = 0.880)
8-sentence chunks	-0.007 (p = 0.572)	-0.005 (p = 0.265)	+0.060 (p = 0.003)	-0.009 (p = 0.936)
top-2 chunks	+0.013 (p = 0.471)	+0.013 (p = 0.108)	-0.058 (p < 0.001)	-0.004 (p = 0.699)
top-5 chunks	+0.005 (p = 0.584)	-0.001 (p = 0.656)	+0.072 (p < 0.001)	-0.032 (p = 0.184)
BM25 retrieval	-0.001 (p = 0.917)	+0.003 (p = 0.933)	+0.028 (p = 0.177)	-0.002 (p = 0.767)
citations not required	+0.021 (p = 0.117)	-0.010 (p = 0.091)	+0.019 (p = 0.314)	—

Negative UFR and CR differences and positive coverage and citation differences favor the variant. Base: 5-sentence chunks, top-3, dense retrieval, citations required.

Question-answering pilot (three questions per course)

Condition	Question	n	Abstained	UFR	CR	Supported by clinician's instructions	Words
QA-E0	all	330	23%	0.429	0.032	16%	31
QA-E0	follow up	110	15%	0.309	0.018	15%	30
QA-E0	medications	110	7%	0.272	0.055	18%	41
QA-E0	warning signs	110	46%	0.893	0.017	13%	20
QA-E1	all	330	33%	0.307	0.032	15%	25
QA-E1	follow up	110	24%	0.246	0.008	15%	24
QA-E1	medications	110	10%	0.212	0.054	15%	36
QA-E1	warning signs	110	65%	0.679	0.026	15%	14

QA-E0 answers from the full hospital course; QA-E1 from the three chunks retrieved with the question. UFR and CR are computed on answered questions only.

Hypotheses: verdicts across both studies

Hypothesis	Pilot (MTSamples, GPT-4o-mini)	Main study (MIMIC-IV, local model)	Verdict
H1: unsupported content is measurable, and the judge agrees with experts better than chance	UFR 0.811 under a judge with kappa 0.06 (chance level)	UFR 0.137 (judge-estimated; verbatim-extraction reference 0.058); judge kappa 0.33	First half supported; second half supported only for the selected judge, moderately
H2: RAG reduces unsupported and contradicted statements	CR down 24% ($p = 0.011$, Holm $p = 0.135$); UFR unchanged	UFR down 45 percent ($p < 0.001$); CR unchanged ($p = 0.985$)	Supported for unsupported facts on hospital courses; not for contradictions
H3: verification adds to RAG	E3 vs E1 UFR $p < 0.001$	Selected judge: E3 = E1 ($p = 0.212$); second judge: 0.079 vs 0.135 ($p < 0.001$)	Supported in the pilot; judge-dependent in the main study

Paired two-sided Wilcoxon tests on the same documents. The main-study judge is DeBERTa-v3-large NLI selected against 1,781 expert-annotated sentences; the second judge is Bespoke-MiniCheck-7B.

Thesis writing status

- Complete second draft generated from the result files: 162 pages in CSUN format (1-inch margins, 12-pt Times New Roman, double-spaced, roman-numbered preliminary pages, arabic body); signature page with the full committee
- Nine chapters: 1 Introduction, 2 Literature Review, 3 Data, 4 Methodology, 5 Pilot Study on MTSamples, 6 Selecting a Valid Judge, 7 Main Study on MIMIC-IV, 8 Discussion, 9 Conclusion; IEEE references with DOIs; Appendices A to E (prompts, per-document results, worked examples, repository, threshold sweep)
- Project status: all experiments promised in the proposal are complete (judge validation and selection, five conditions on MTSamples and on 110 MIMIC-IV hospital courses, coverage against the clinician's instructions, citation accuracy, six retrieval ablations, question-answering pilot), plus a second-judge robustness check on all 6,588 main-study claims
- Administrative status: committee confirmed (Dr. Wang, chair; Dr. McIlhenny; Dr. Ebrahimi); ETD planning form submitted
- Every table and figure is regenerated by one command from the result files

Revisions in the second draft

- Committee members added to the signature page; acknowledgments rewritten
- Statistics corrected: the retrieval-ablation effects on faithfulness are now reported with Holm-adjusted p-values and do not survive the correction (coverage effects do); the pilot's Holm family is described correctly as all twenty pairwise tests
- Hypotheses H1 to H3 answered across both studies in Chapter 8; paired tests added under the second judge (verification best under that judge, $p < 0.001$)
- Literature review extended with grounding-specific checkers (MiniCheck), evidence on unreliable factuality metrics (Godbole and Jia 2025; FaithBench 2025) and clinical fact verification at scale (VeriFact 2026)
- Consistency pass: Chapters 1 to 4 brought in line with the main study (scope, data sources, methodology overview, section numbering), the comparison of approaches in Section 8.4 rewritten for both studies, spelling and figure captions corrected
- Manual audit completed in two blind passes on an extended sample (400 items; the rubric was clarified after the first pass and the agreement between the passes is reported): it confirms E1b's reduction against E0 directly and shows the judge understates absolute unsupported shares
- Generator robustness: E0, E1, E1b and E3 re-generated on all 110 courses with Llama-3.1-8B and Mistral-7B and scored by the same judge; the grounding reductions replicate under Llama but not under Mistral, so the thesis states them per generator

Plan to the defense (aligned with CSUN ETD deadlines)

By	Milestone
Oct 9, 2026	ETD Planning Form submitted; committee confirmed (done)
Mid-October	Second draft and progress report submitted; advisor review of the second draft
Oct 23, 2026	Advisor approval; Canvas upload and Turnitin similarity check (20% or lower)
Oct 23 to Nov 6, 2026	Draft to committee (at least two weeks of review); Doodle poll with 1-hour slots over 3 to 4 weeks
Nov 6, 2026	ETD preliminary format review
Nov 16 to 24, 2026	Defense on CSUN Zoom (40 min presentation, 10 min Q&A, 10 min deliberation); slides to advisor one week before
Dec 4, 2026	Final ETD upload after revisions; committee approval in the ETD system

The defense is planned before the December 4 ETD deadline, as the committee approves the thesis inside the ETD system.

Risks and requests

- Judge validity: the selected judge agrees with experts moderately (kappa 0.33); main-study rates are judge-estimated and reported with this error profile, checked by a second judge and a manual audit, and conclusions rest on paired comparisons
- Generator difference: the pilot used GPT-4o-mini on public notes and the main study a 7-billion-parameter local model on MIMIC-IV, so the two studies are compared qualitatively; within the main study the grounding effect replicated under Llama-3.1-8B but not under Mistral-7B, so it is stated per generator
- Schedule: advisor approval, the Turnitin check and two weeks of committee review must all precede the November 6 format review
- Requests to the advisor: review the second draft and approve it for distribution to the committee; advise on the defense window